

**ARCHITECTURE
PORTFOLIO**

**NG AN QI
2012-2017**

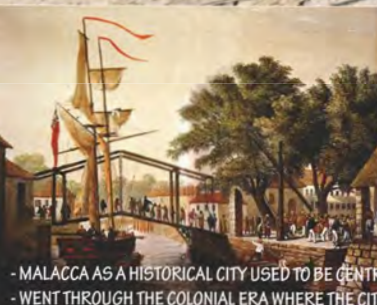
P2: MALACCA VISITOR CENTRE

THIS PROJECT IS TO DESIGN A VISITOR CENTRE BY RESPONDING TO THE SOCIO-CULTURAL AND PHYSICAL CONTEXTS AT MALACCA HISTORICAL SITE
WE ARE REQUIRED TO DESIGN A DOUBLE STOREY BUILDING WITH A MAXIMUM OF 500m² BUILT-UP AREA FOR 6-8 SPACES

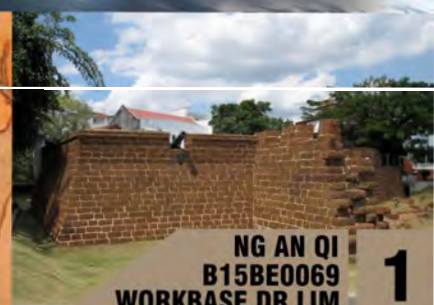


RESTORATION OF MALACCA FORT

TO REVIVE THE OLD MALACCA FORT
TO CONVEY THE HISTORICAL STORY BY THE BUILDING ITSELF
TO BECOME A CENTRE OF VALUABLE HISTORICAL INFORMATIONS

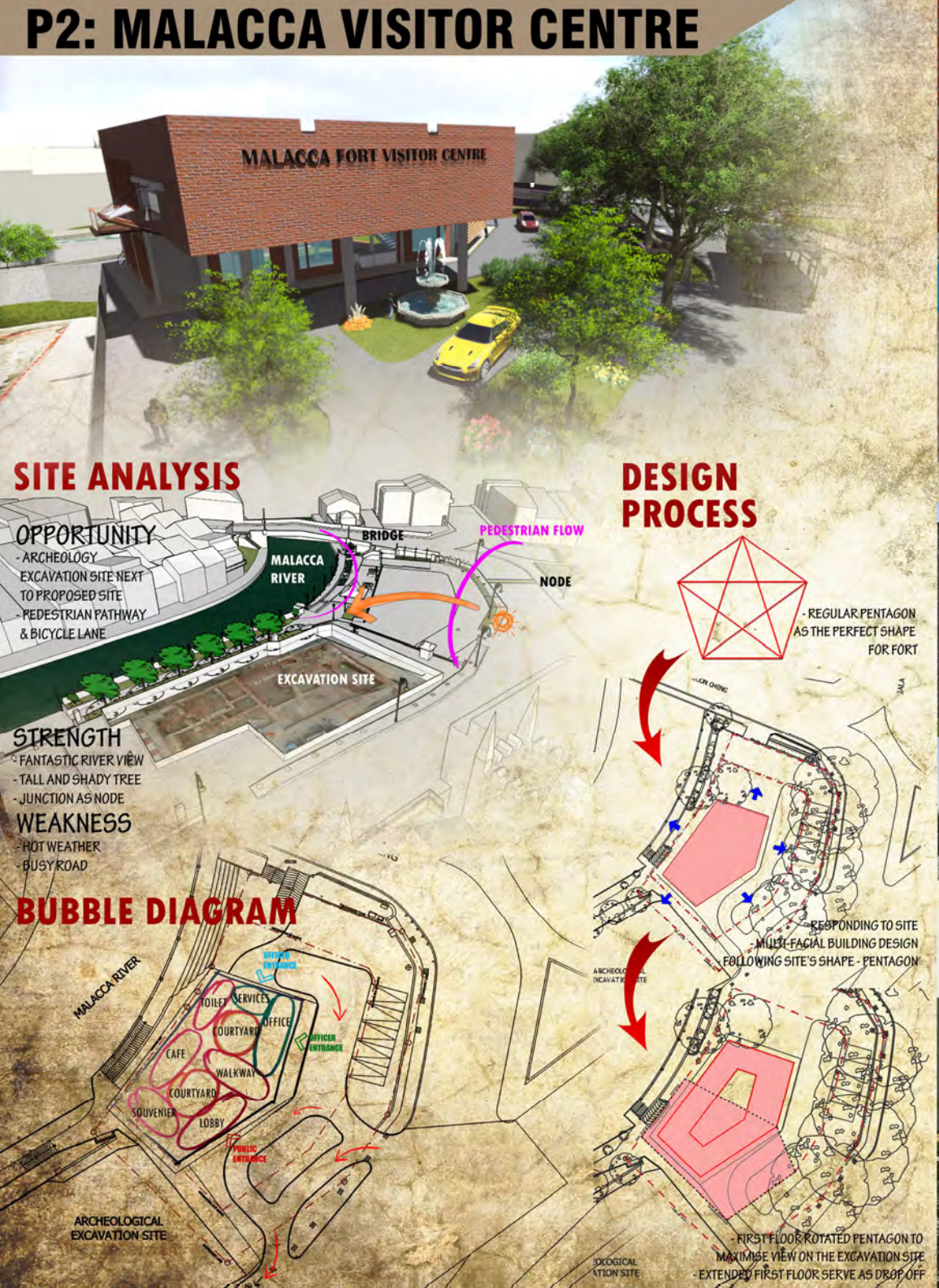


- MALACCA AS A HISTORICAL CITY USED TO BE CENTRAL OF TRADE AREA
- WENT THROUGH THE COLONIAL ERA WHERE THE CITY WAS ATTACKED



- VICTORIA BASTION - REMAININGS OF MALACCA FORT JUST NEXT TO THE SITE

P2: MALACCA VISITOR CENTRE



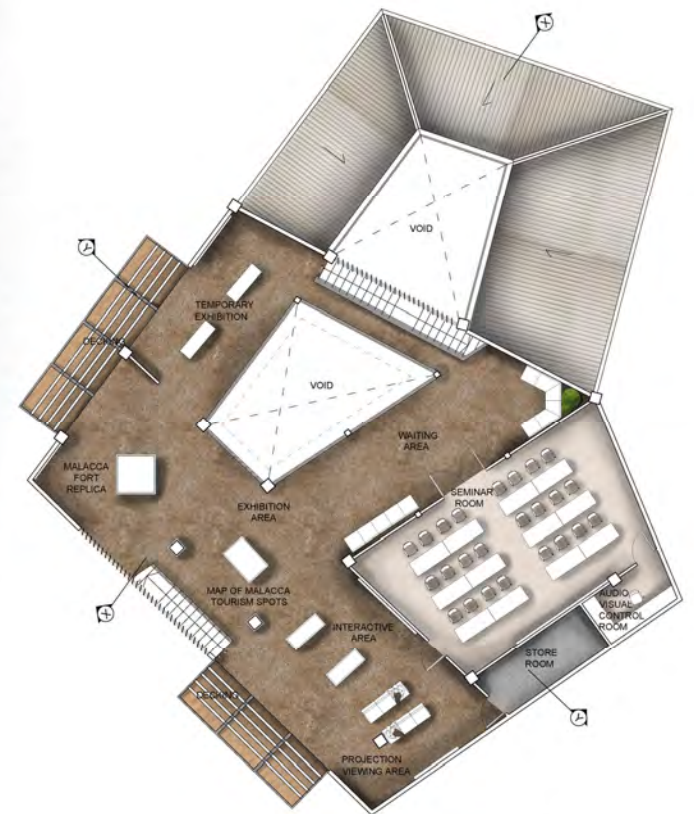
P2: MALACCA VISITOR CENTRE



P3: MELAKA VISITOR CENTRE



GROUND FLOOR PLAN
SCALE 1:100



FIRST FLOOR PLAN
SCALE 1:100

P2: MALACCA VISITOR CENTRE



NORTHEAST ELEVATION
SCALE 1:100



SOUTHEAST ELEVATION
SCALE 1:100



SOUTHWEST ELEVATION
SCALE 1:100



NORTHWEST ELEVATION
SCALE 1:100

P2: MALACCA VISITOR CENTRE

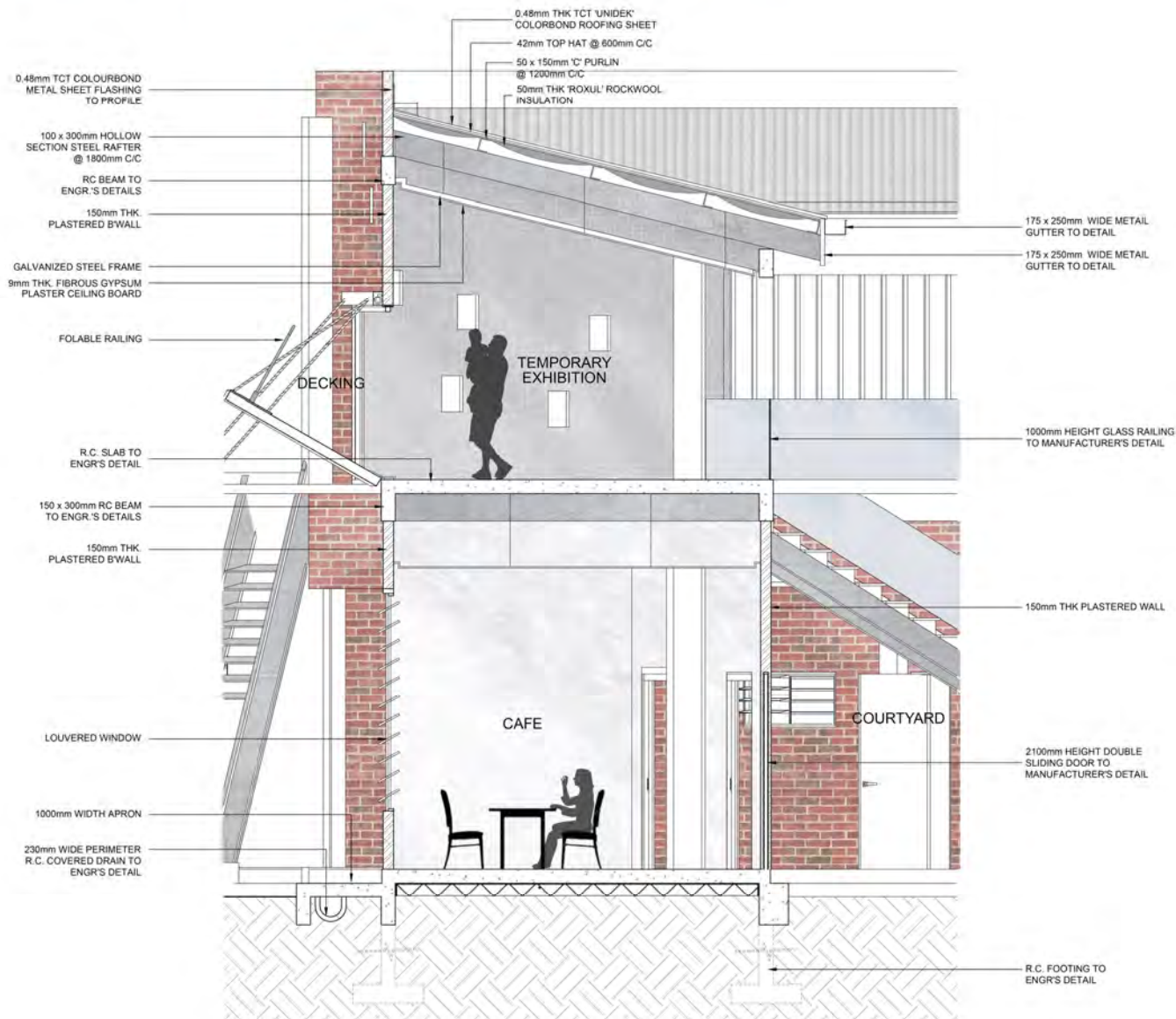


SECTION X-X
SCALE 1:100

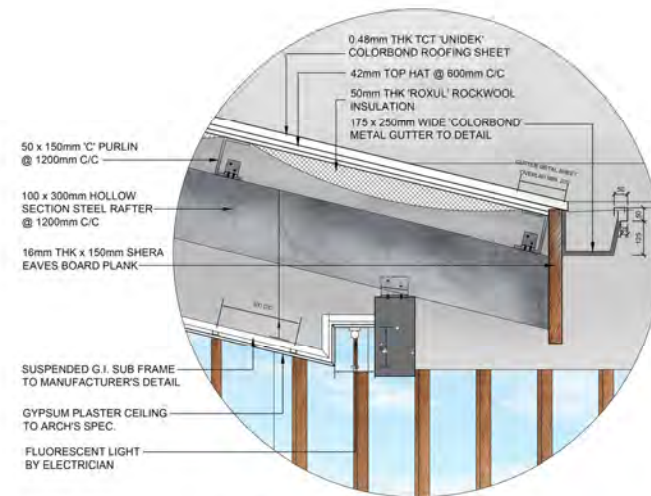


SECTION Y-Y
SCALE 1:100

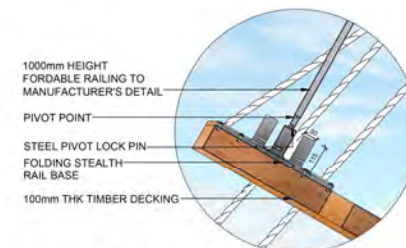
P2: MALACCA VISITOR CENTRE



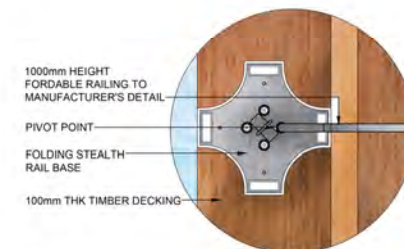
SECTIONAL DETAIL
SCALE 1:25



ROOF DETAIL
SCALE 1:10



FOLDABLE RAILING DETAIL
SCALE 1:10



FOLDABLE RAILING DETAIL
SCALE 1:10

P2: MALACCA VISITOR CENTRE



PERSPECTIVE OF RIVERSIDE



STAIRCASE TO EXHIBITION HALL



PERSPECTIVE OF COURTYARD



NIGHT VIEW



PERSPECTIVE FROM RIVER



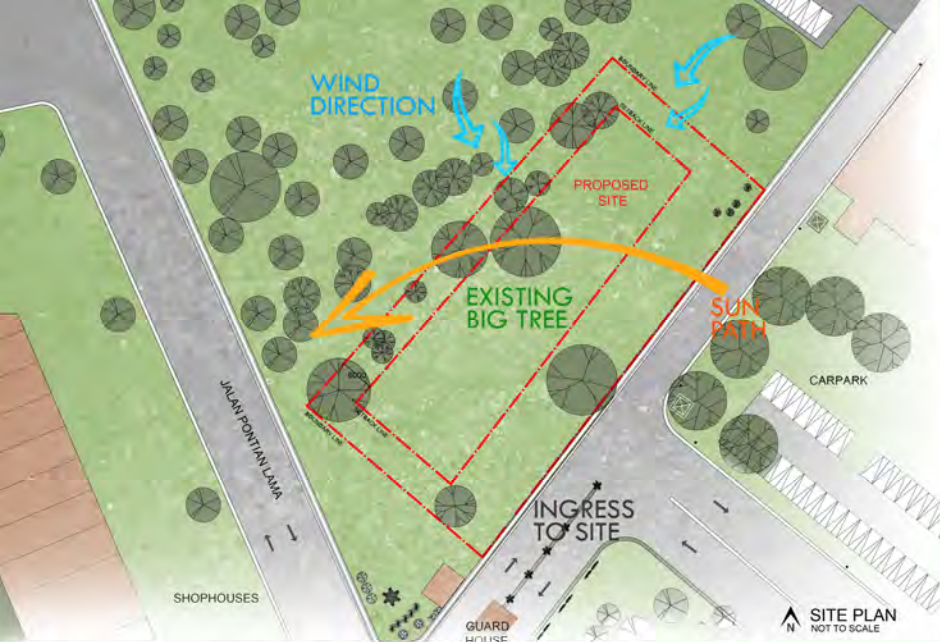
PERSPECTIVE FROM BICYCLE LANE

WAVY BREEZE

ORGANIC FORM OF BUILDING INSPIRED BY NATURE WHILE
UTILIZE THE NATURAL FORCES TO CREATE A SUSTAINABLE
GREEN BUILDING WHICH IS SELF COOLING BY PASSIVE
VENTILATION AND THE POOL DESIGNED TO BE
SURROUNDING THE BUILDING



THE BRIEF CALLS FOR THE DESIGN OF "IBM SMARTER
CITIES TRAINING CENTRE, SKUDAI
INNOVATION VALLEY" LOCATED IN UTM JOHOR
BAHRU CAMPUS. THE BUILDING IS AN ACADEMIS
CENTRE, COMBINING THE FUNCTIONS OF AN IT
TRAINING CENTRE AND ADMINISTRATIVE OFFICE FOR
PROMOTING SMARTER CITIES, WHICH IS ONE OF THE
KEY SOLUTIONS OFFERED BY THE COMPANY. THE
BUILDING SHALL PORTRAY ITSELF AS AN
ENVIRONMENTALLY SUSTAINABLE ARCHITECTURE FOR
THIS CLIMATE AND SITE CONTEXT.



SITE ANALYSIS & SYNTHESIS

STRENGTH

- BIG SHADY TREES AT SITE
- EXISTING GUARDHOUSE AT JUNCTION

WEAKNESS

- ELONGATED FACING SOUTHEAST SITE

OPPORTUNITY

- SHORT FRONTAGE FACING ROAD AS PUBLIC ATTRACTION
- WINDY SITE WITH NICE VENTILATION

THREAT

- BUSY ROAD WITH LOUD TRAFFIC NOISE

THE EXISTING BIG TREES AT SITE SHOULD BE PRESERVED AND RESPECTED WHILE FULLY UTILIZE IT TO PROVIDE SHADE

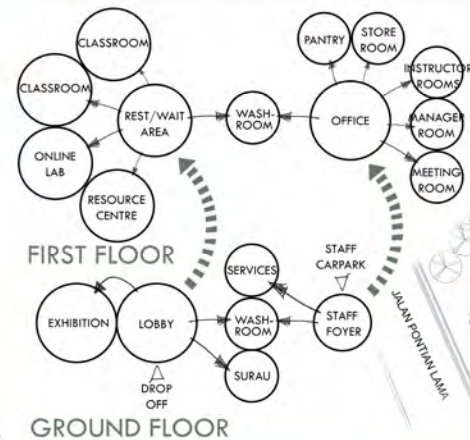
BUILD INTERESTING FACADE TO ATTRACT PUBLIC FROM MAIN ROAD

BY FOLLOWING TROPICAL CLIMATE AND RESPECTING THE SITE EXISTING BIG TREE, THE BUILDING IS DESIGNED TO BE 3 BLOCKS WHICH ENCOURAGES CROSS VENTILATION WHILE SURROUNDING THE TREES AT THE SITE.

THE ORGANIC FORM OF THE BUILDING IS DESIGNED SO AS TO RESPECT THE ORGANIC NATURE WHERE THE FACADE ALSO SERVE AS SUNSHADING DEVICE FOR THE BUILDING.



SPATIAL ARRANGEMENT

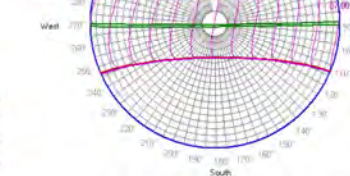


SUN SHADING CONSIDERATIONS

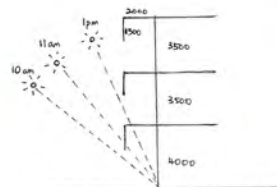
LOCATION: JOHOR
LATITUDE: 1.48° N
LONGITUDE: 103.73° E

SUN PATH SHOWN:

- JUNE 22
- MARCH 22
- SEP 22
- JAN 1
- DEC 21
- DEC 22



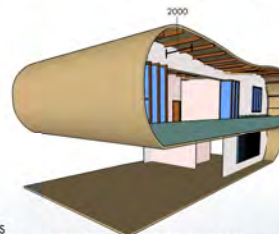
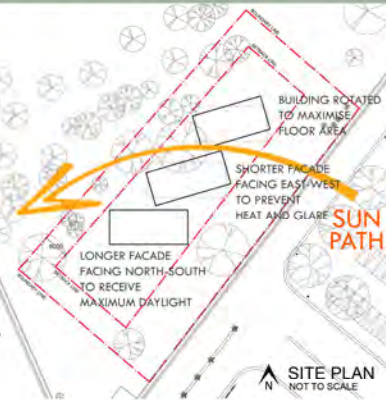
SUN PATH DIAGRAM



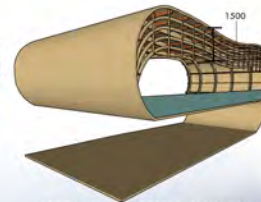
SECTIONAL SKETCH

SUN PATH IN JOHOR DIFFERS ACCORDING TO MONTHS AND HOURS, DURING DAYS IN JUNE, THE SUN ANGLE CAN GO DOWN TO 40° ANGLE FROM GROUND AT 10am

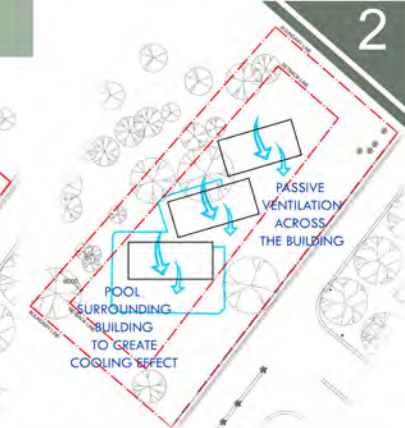
PROPER SUN SHADING DEVICE SHOULD BE INSTALLED TO BLOCK HEAT AND GLARE UP TO THE LOWEST POINT THE SUNLIGHT CAN SHINE IN THE BUILDING



FACADE IS DESIGNED TO HAVE 2000mm OVERHANG TO BLOCK DIRECT SUNLIGHT GLARE AND HEAT



1500mm OF EGG CRATE SUN SHADING DEVICE IS DESIGNED TO BLOCK JUNE AND DECEMBER SUNLIGHT SHINING INTO THE BUILDING



VENTILATION

10mm THK EXTRUDED ALUMINIUM LOUVRE BLADES CONNECTOR

10mm THK GLASS LOUVRE BLADES

50mm THK TIMBER WINDOW FRAME

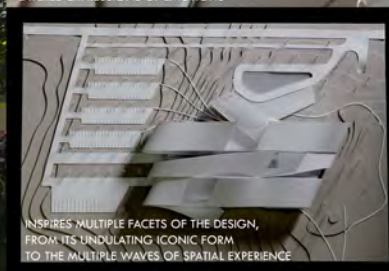


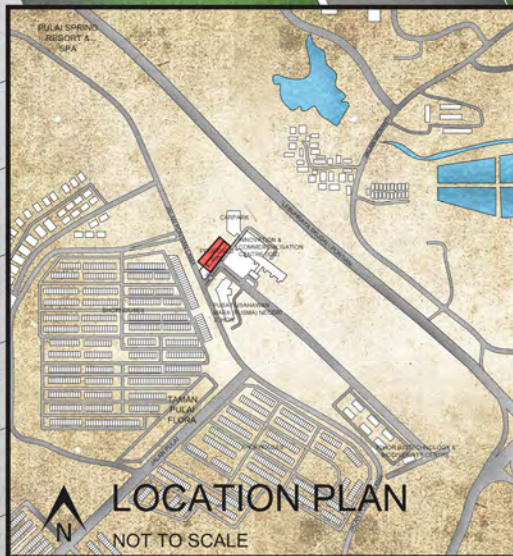
VERTICAL LOUVRE DETAIL

VERTICAL LOUVRES ARE DESIGNED IN A V-SHAPED MANNER TO PREVENT RAINWATER FROM SPLASHING INTO THE BUILDING BUT ALLOWING DAYLIGHT TO ENTER THE SPACES AND ALSO ENCOURAGE VENTILATION TO FLOW INTO THE BUILDING

PRECEDENCE

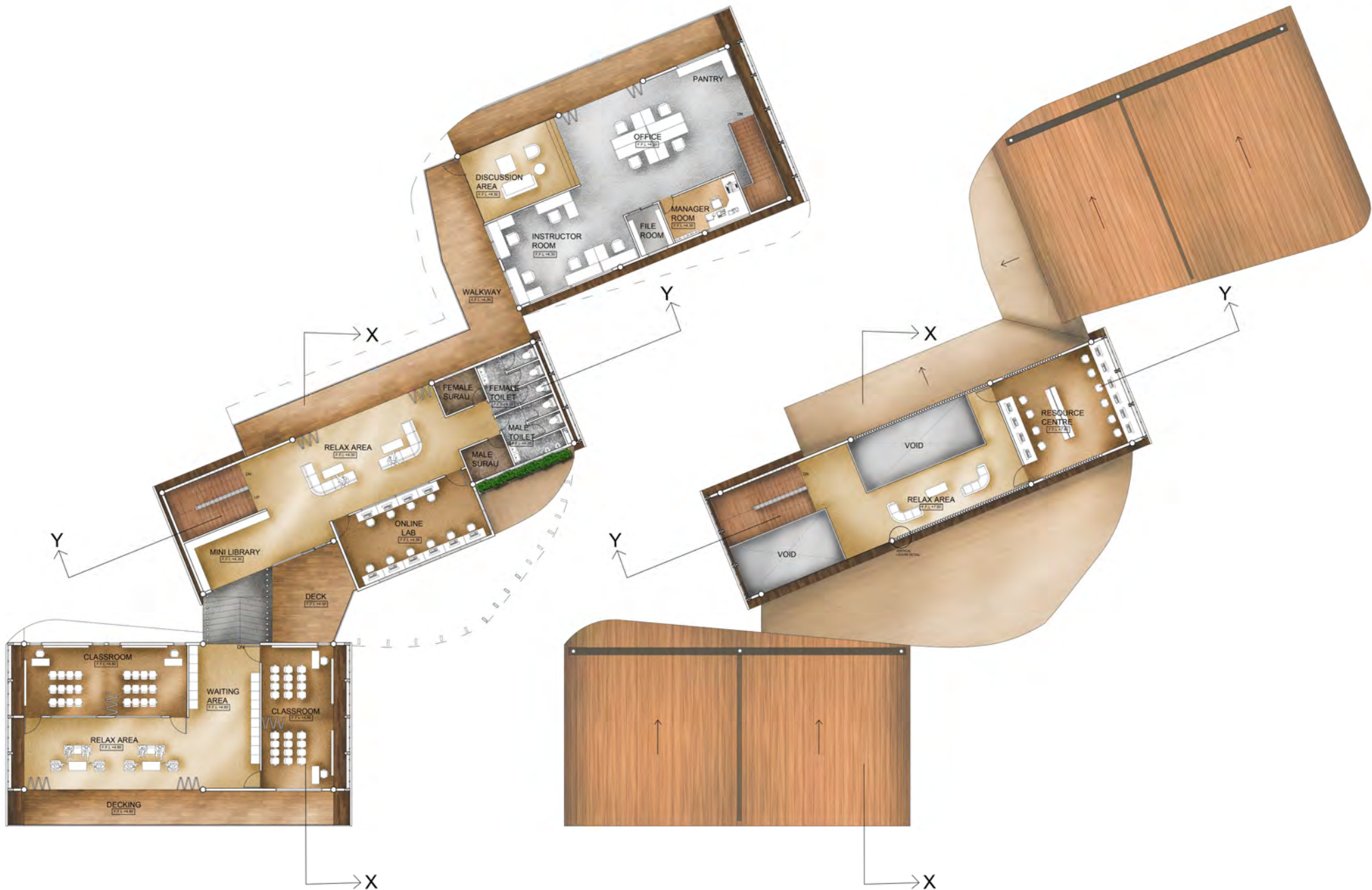
BOLLYWOOD MUSEUM
BY YAZDANI STUDIO OF CANNON DESIGN





P2: IBM SMARTER CITIES TRAINING CENTRE





FIRST FLOOR PLAN

SCALE 1:100



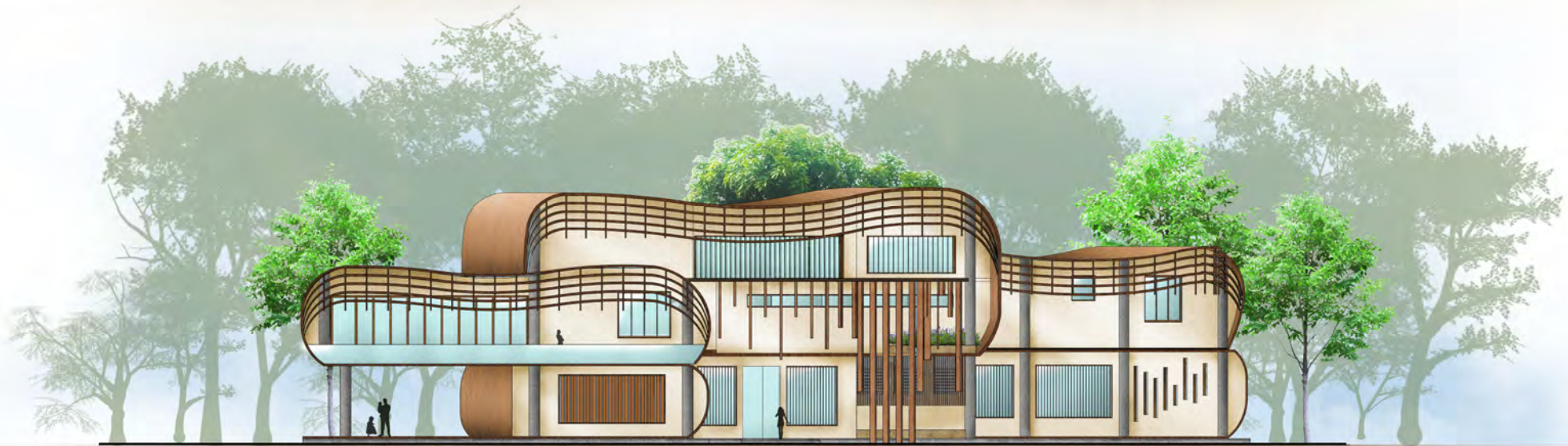
SECOND FLOOR PLAN

SCALE 1:100



NORTH ELEVATION

SCALE 1:100



SOUTH ELEVATION

SCALE 1:100



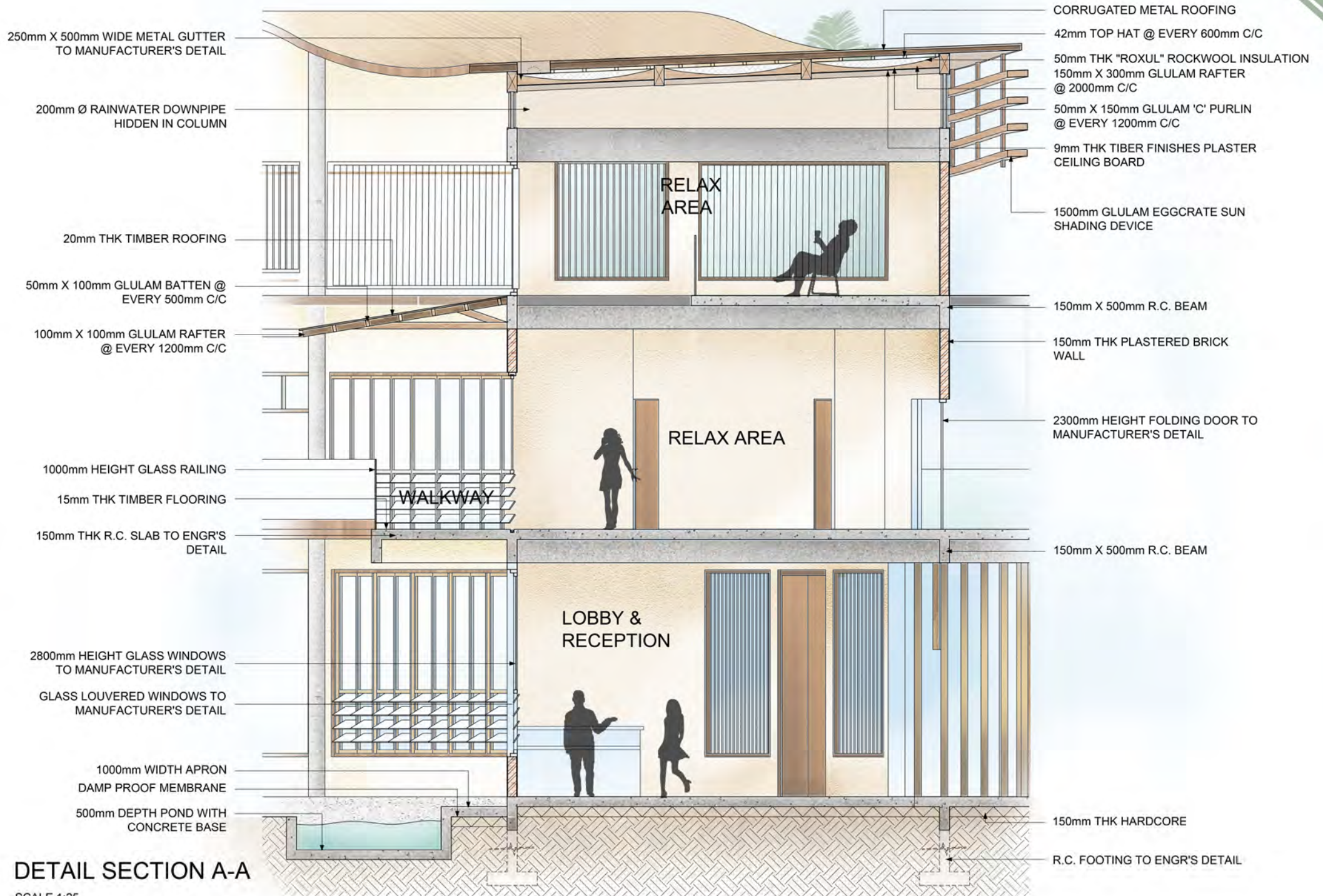
SECTION X-X

SCALE 1:100



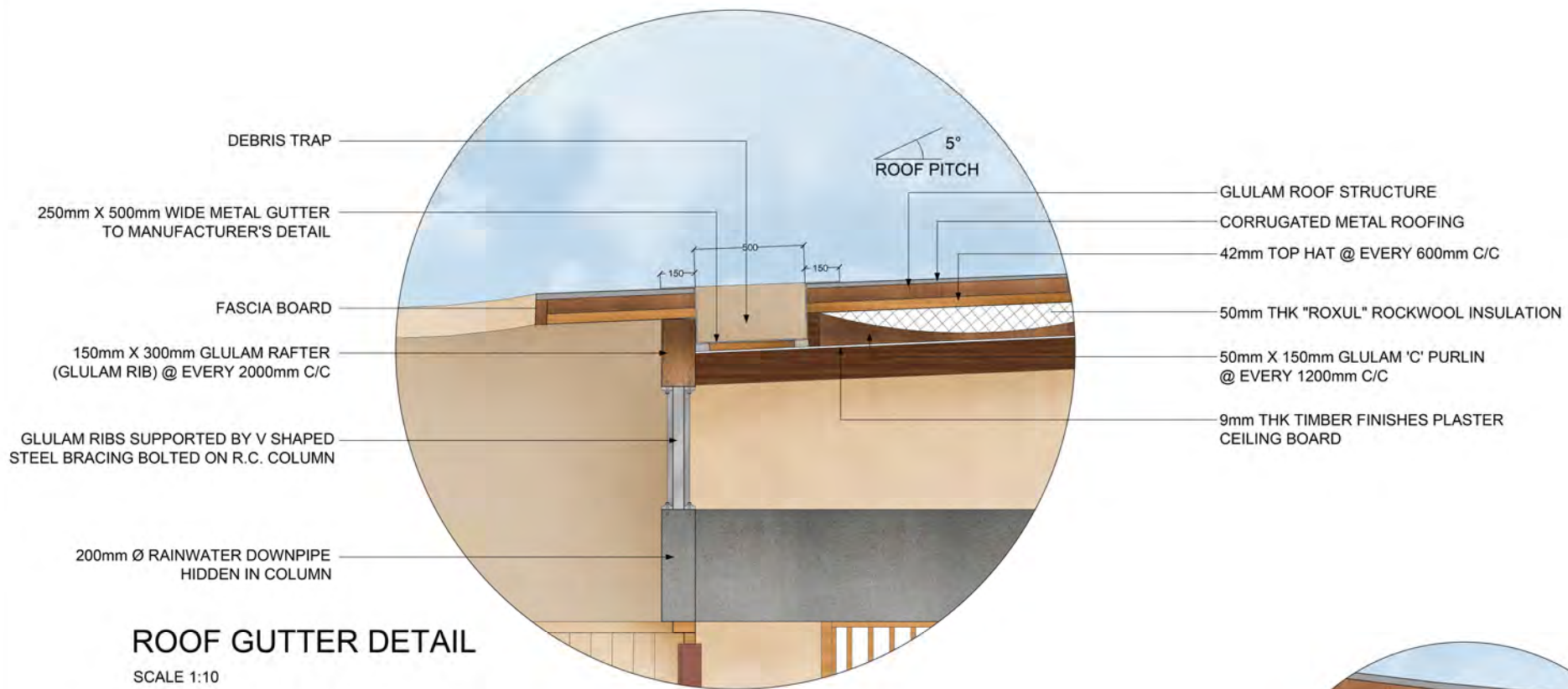
SECTION Y-Y

SCALE 1:100



DETAIL SECTION A-A

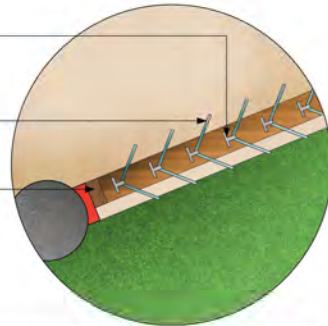
SCALE 1:25



10mm THK EXTRUDED ALUMINIUM LOUVRE BLADES CONNECTOR

10mm THK GLASS LOUVRE BLADES

50mm THK TIMBER WINDOW FRAME



VERTICAL LOUVRE DETAIL

SCALE 1:10

GLULAM RIB

STEEL FLITCH PLATES

50mm X 100mm STEEL HOLLOW SECTION BOLTS

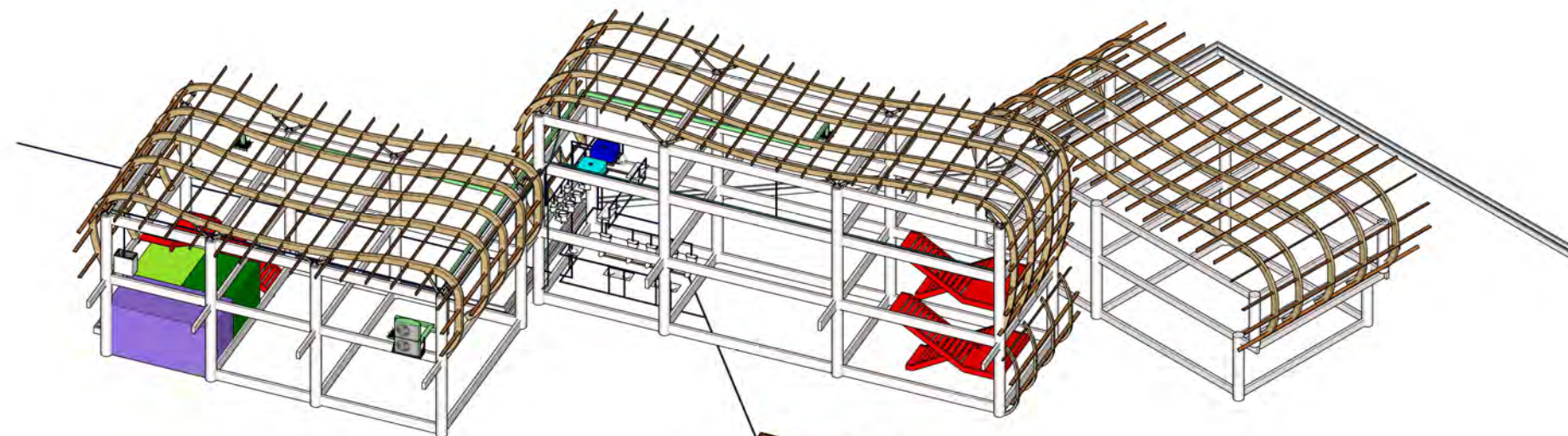
PIN CONNECTION

R.C. COLUMN



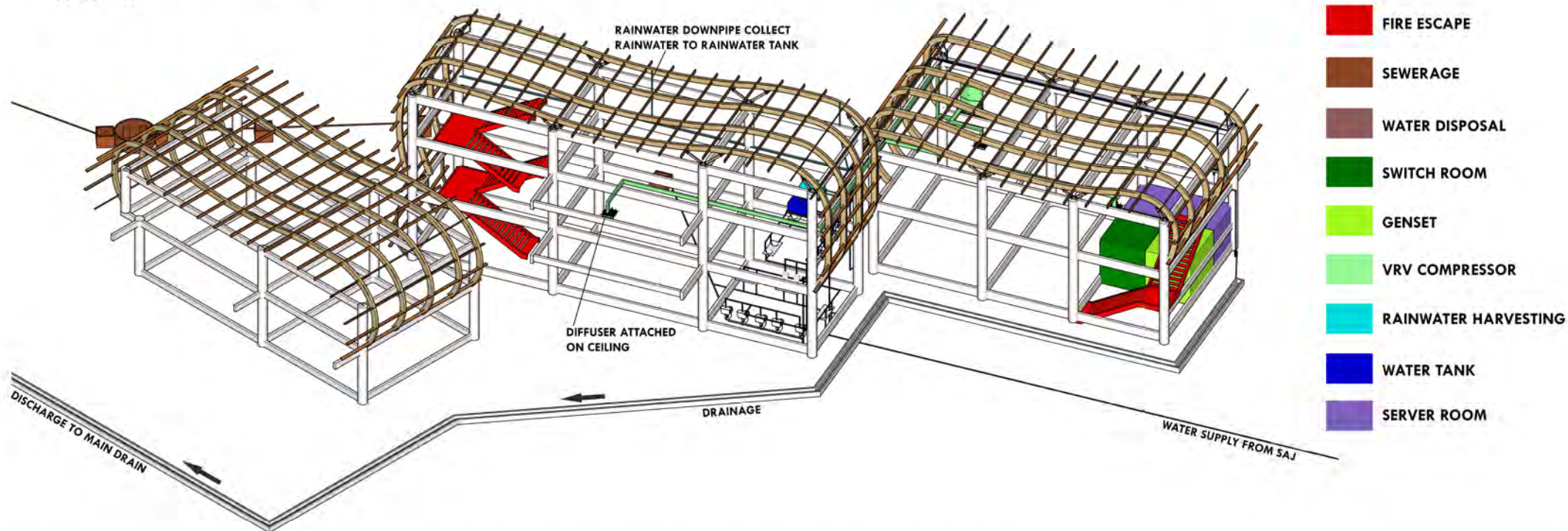
'V' SHAPED BRACING CONNECTION DETAIL

SCALE 1:10



ISOMETRIC SERVICES AND STRUCTURAL DIAGRAM

SCALE 1:100



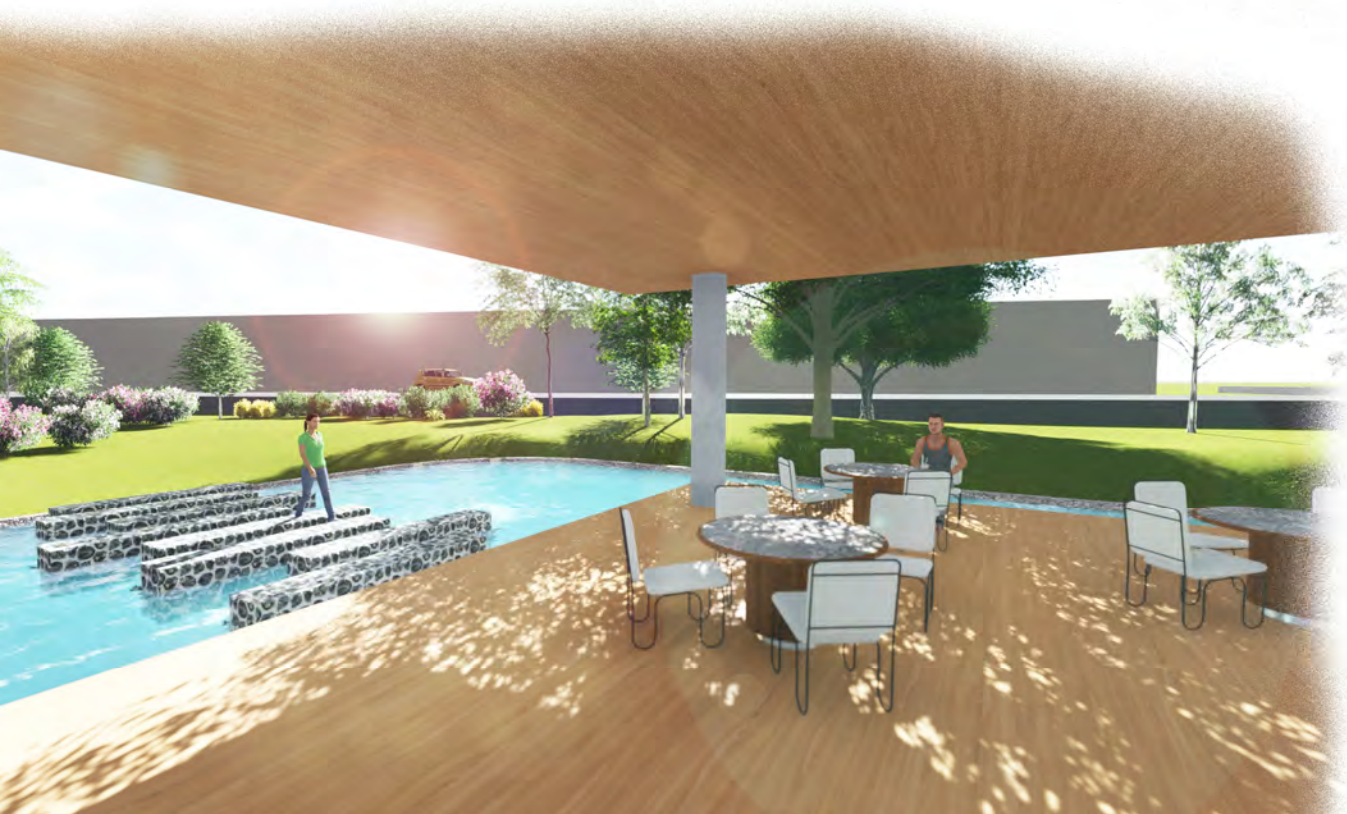
SUNSET VIEW OF BUILDING
- VARIOUS ACTIVITY TAKING PLACE ON DIFFERENT LEVELS OF
THE BUILDING WHILE HAVING MAXIMUM INTERACTION
WITH THE NATURE



LOBBY & RECEPTION

- LOBBY IS DESIGNED WITH FULL HEIGHT OPENINGS ON BOTH SIDES TO ENCOURAGE NATURAL DAYLIGHT AND VENTILATION
- LOBBY OPENS UP VIEW TO THE EXISTING BIG TREE

- REST & WAITING AREA FOR STUDENTS**
- STUDENTS CAN RELAX AND STUDY AT THE RESTING AREA DURING FREE TIME WHILE ENJOYING NATURAL VIEW
 - STUDENTS CAN WAIT AND DISCUSS AT THE WAITING AREA DURING CLASSES



OPEN LEISURE AREA

- A PLACE FOR PEOPLE TO RELAX, ENJOY THE VIEW AND THE COOL BREEZING AIR
- A MARBLE WALKWAY ON THE POOL WHICH ENABLES PEDESTRIAN TO ENTER THE BUILDING

BACK FACADE OF THE BUILDING

- RESPECTING AND APPRECIATING THE EXISTING BIG TREE, A LOT OF OPENINGS AND WALKWAY ARE DESIGNED SURROUNDING THE TREE WHICH ALSO PROVIDES GREAT SHADE TO THE INTERIOR SPACES.



VIEW FROM JALAN PONTIAN LAMA
- INTERESTING FACADE & POND THAT PORTRAYS
A LIVELY TRAINING CENTRE TO PUBLIC

SEM 1 2016/2017

3rd Year 1st Sem



NG AN QI
B15BE0069
WORKBASE DR LIM.Y.L.

BUILDING PROGRAMME

Food Cultural Centre
- Bridge



- Provide food for Malaysian that works in Singapore
- Introduce Malaysia Culture by food to tourists and foreigner
- Revive movement of KTM by encouraging pedestrian walk
- Connect people by food and movement (walking)

PRECEDENCE



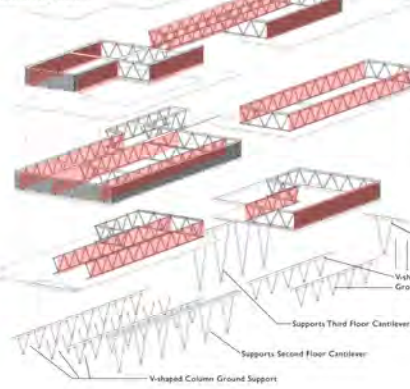
Mashout Performing Arts Center
- 5 performance spaces linked through a series of transparent hallways and classrooms
- One continuous horizontal layer connects entire building



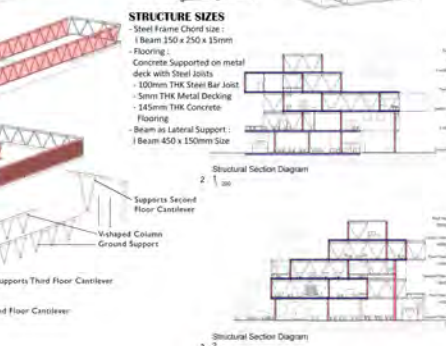
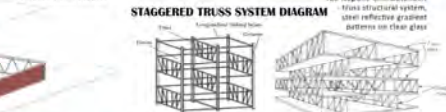
De Brug (Unilever Bestfoods office)
- Bridge over an existing historical factory
- Glass and glass material 20 meters above ground
- Transparency design give communicative character

STAGGERED TRUSS SYSTEM

- Structural Steel Framing used for a large column-free area
- Interaction of Floors, Trusses and Columns to form strength and rigidity
- Lightweight Steel Structure
- Less Foundation Formwork
- Fire resistance by using spray-on protection, the applied thickness can be kept to a minimum due to the compact nature of the truss elements
- Steel framing member as Vertical Load Transfer
- Concrete floor as Horizontal Load Transfer

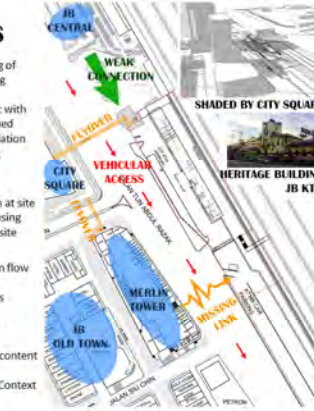


CASE STUDIES

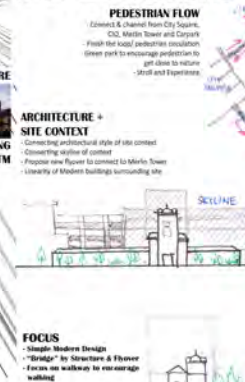


SITE ANALYSIS

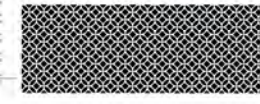
- STRENGTH:**
- Shadow casting of High-rise during Evening Time
 - One way traffic with properly-planned vehicular circulation
 - Proper existing facilities
- WEAKNESS:**
- Less vegetation at site
 - High traffic causing jam in front of site
- OPPORTUNITY:**
- High pedestrian flow at site
 - Strategic Site as Connection of Singapore and Malaysia
 - High historical content of site
 - Linkages from Context by Flyover



IDEATION: BRIDGING

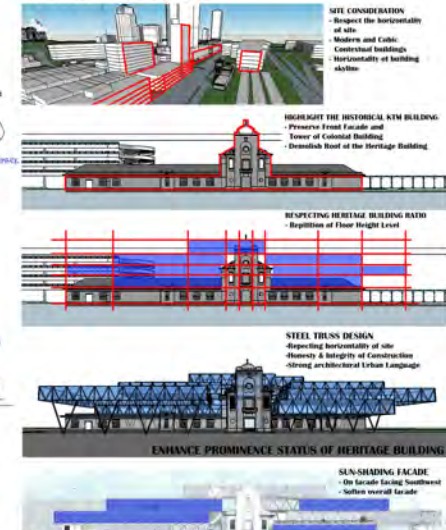


FACADE DESIGN



- Soften industrial-like appearance
- Sun-shading facade design facing southeast
- Art deco Pattern facade

DESIGN PROCESS



URBAN-BRIDGE PROMENADE

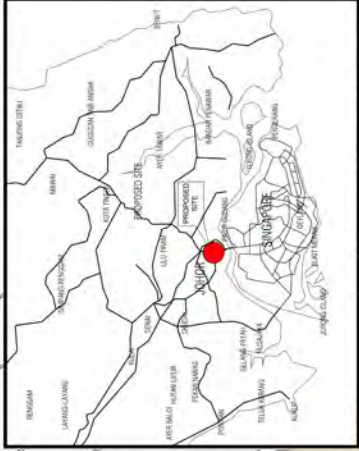
NG AN QI

B15BE0069

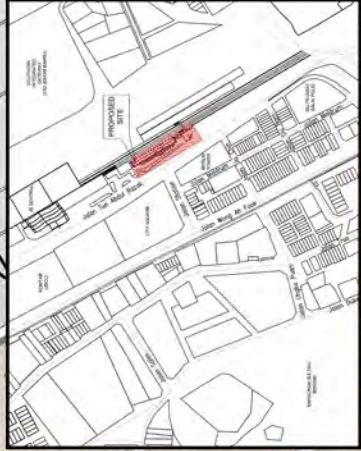
WORKBASE DR LIM Y.L.

Food Hub Centre

The project requires exploration on local heritage in producing a community-oriented design. The building demonstrates new intervention on adaptive re-use development at Johor Bahru town. Serve as food hub centre, the building provide Food Marketing to local and tourists.



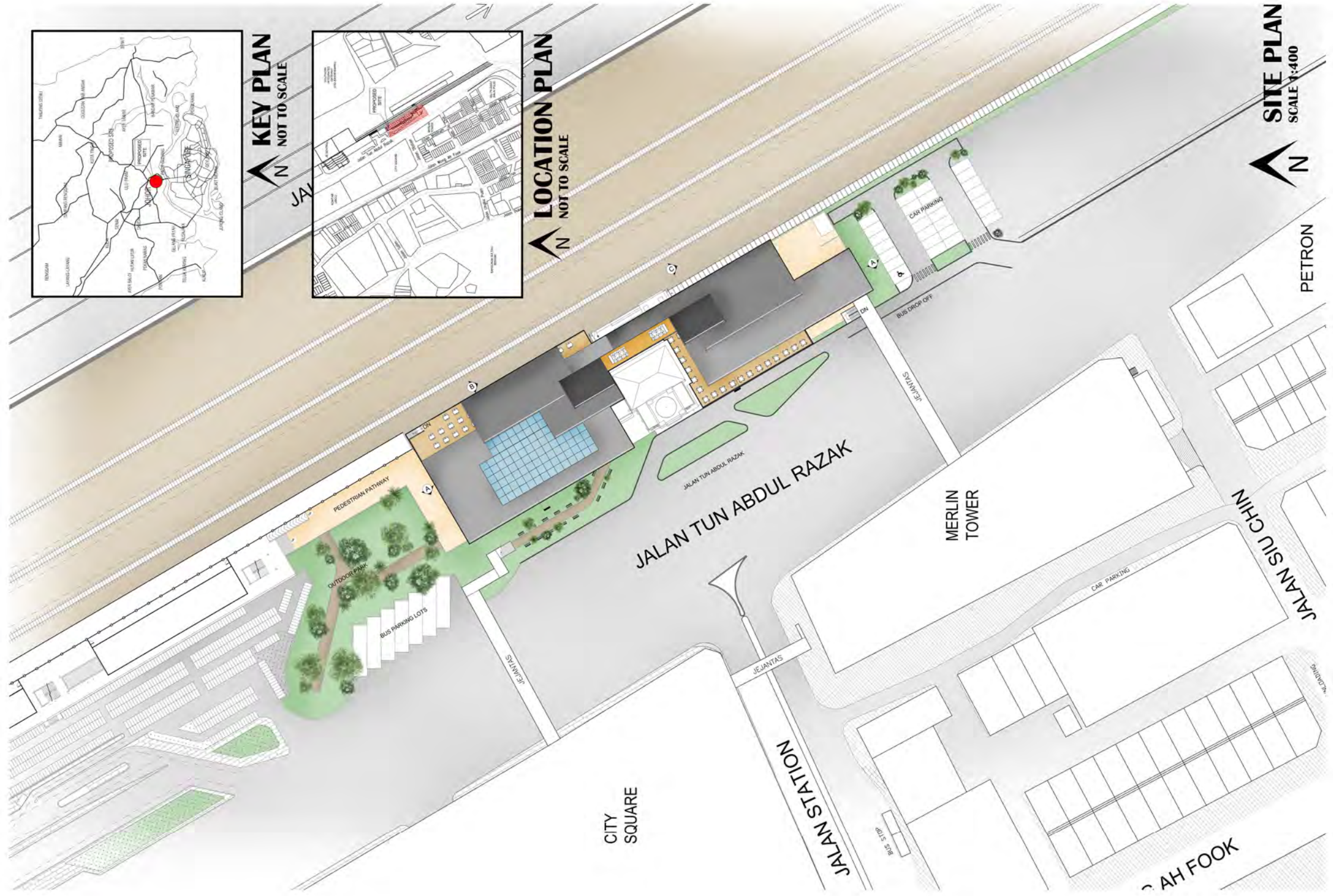
KEY PLAN
NOT TO SCALE

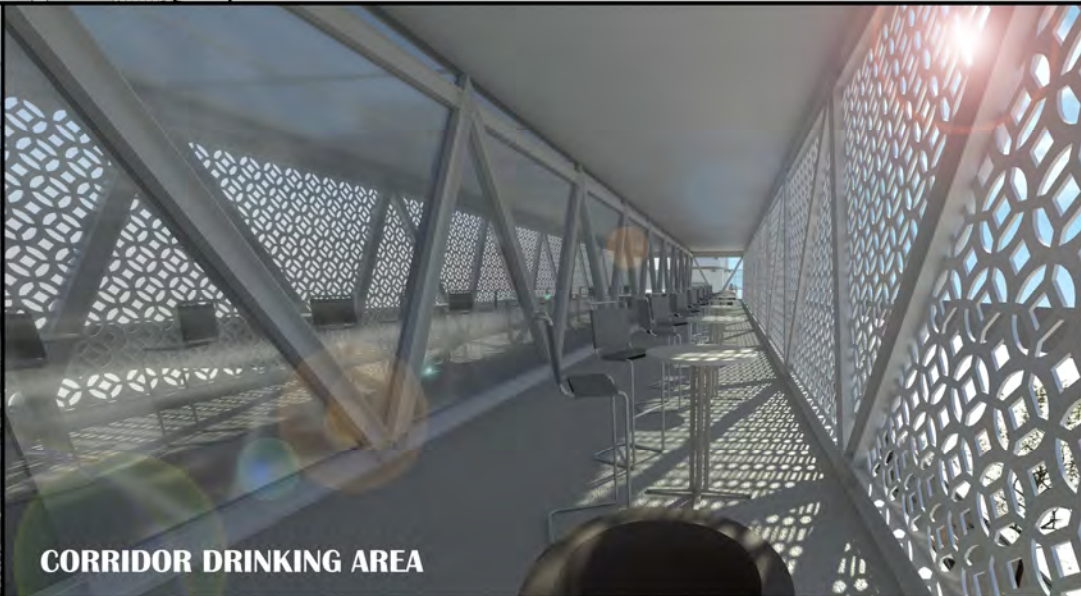
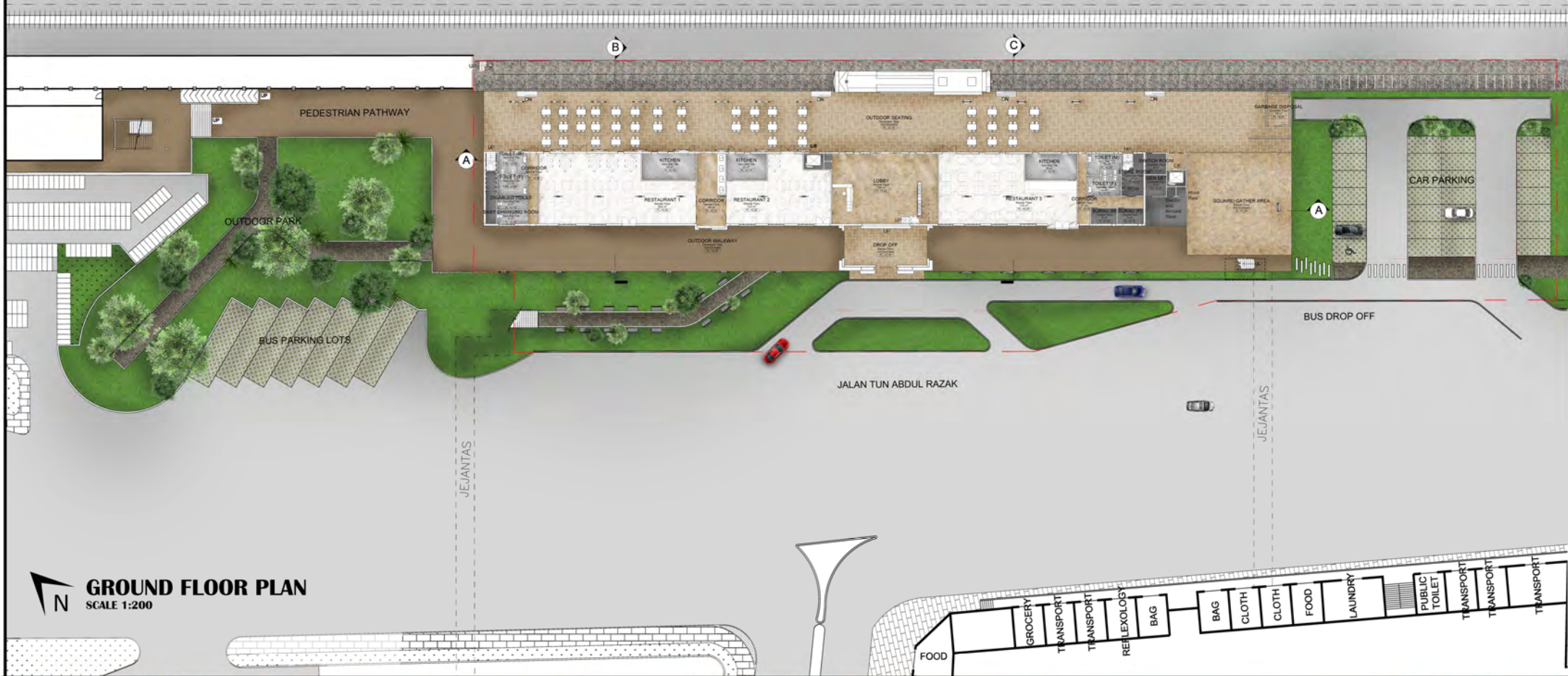


LOCATION PLAN
NOT TO SCALE



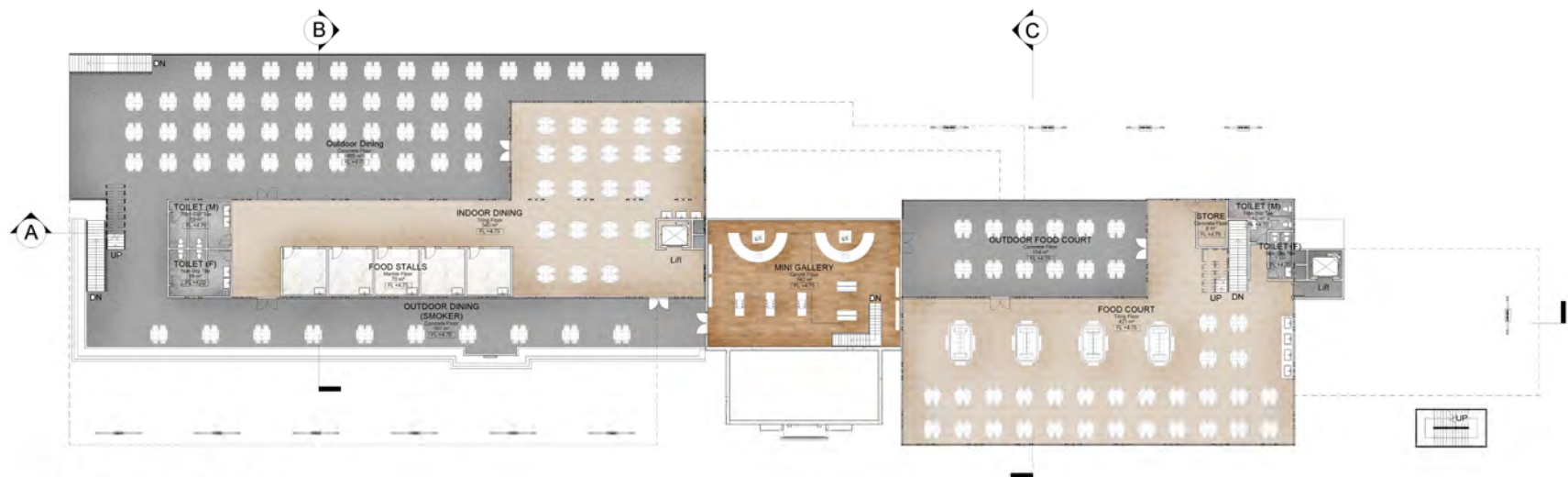
SITE PLAN
SCALE 1:400



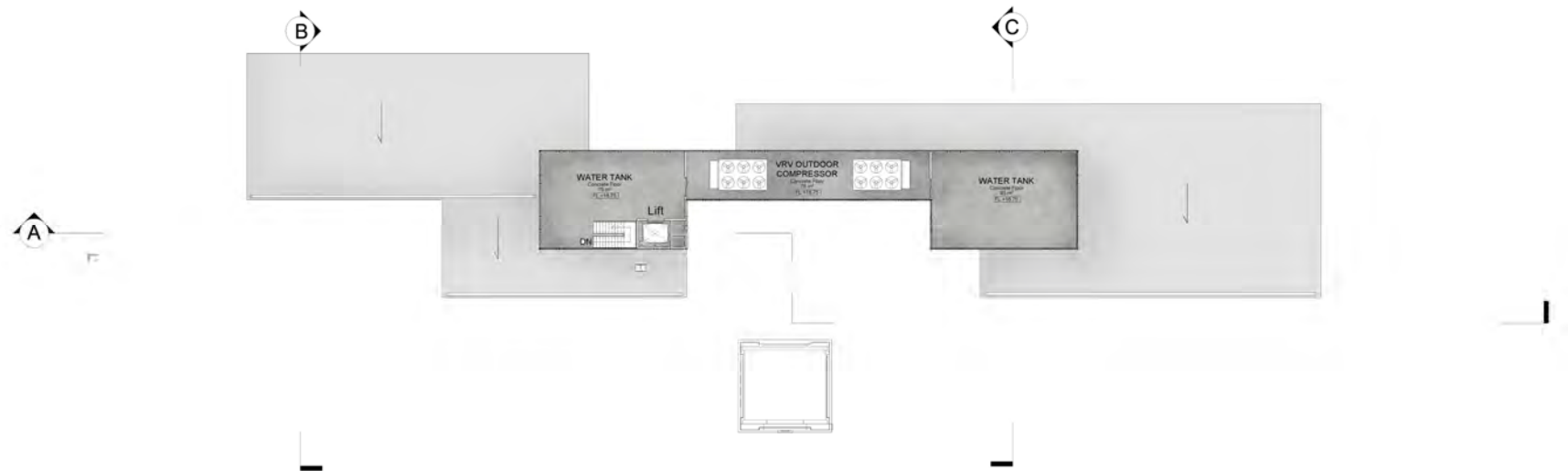




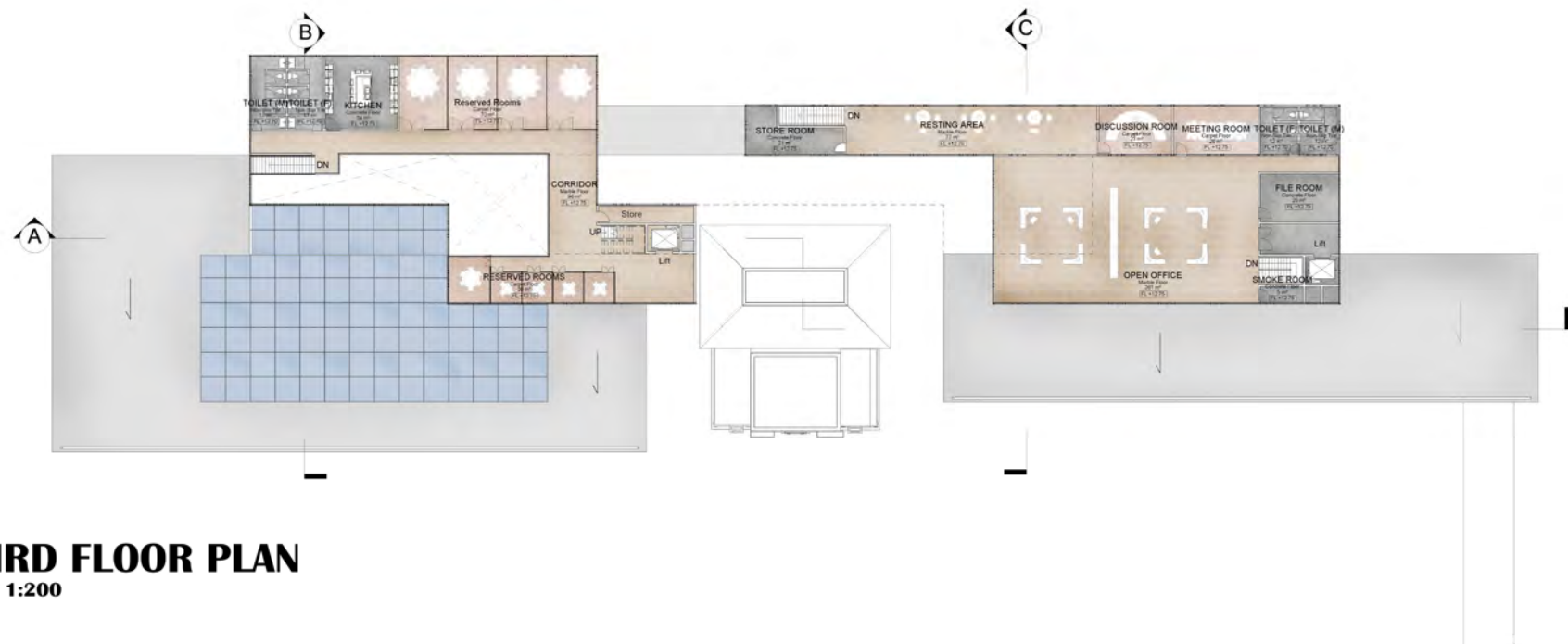
SECOND FLOOR PLAN
SCALE 1:200



FIRST FLOOR PLAN
SCALE 1:200



FOURTH FLOOR PLAN
SCALE 1:200



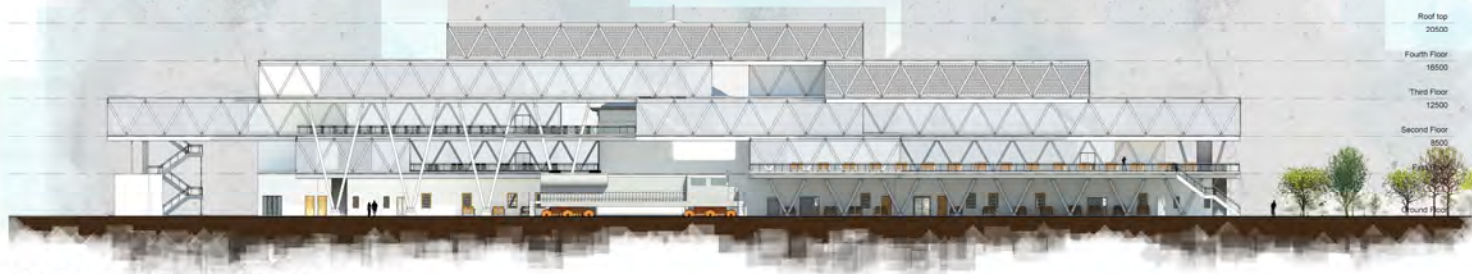
THIRD FLOOR PLAN
SCALE 1:200



SOUTHWEST ELEVATION
SCALE 1:200



SOUTHEAST ELEVATION
SCALE 1:200



NORTHEAST ELEVATION
SCALE 1:200



NORTHWEST ELEVATION
SCALE 1:200



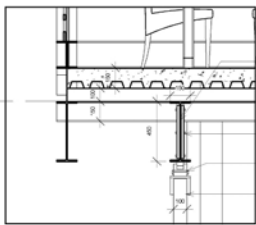
SECTION A-A
SCALE 1:200



SECTION B-B
SCALE 1:200

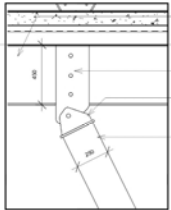


SECTION C-C
SCALE 1:200



450mm x 150mm Steel I Beam
First Floor
4500
Bracket Plate
Gusset Plate
100mm x 250mm V-Shaped Steel Column

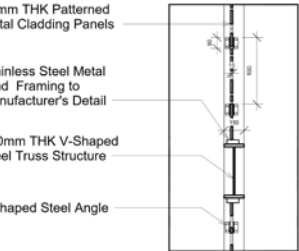
V COLUMN CONNECTION
SCALE 1:20



450mm x 150mm Steel I Beam
First Floor
4500
Bracket Plate
Gusset Plate
100mm x 250mm V-Shaped Steel Column
200mm x 50mm Stainless Steel Scupper Drain to Engr's Detail

Sheet Metal Coping

15mm THK Patterned Metal Cladding Panels

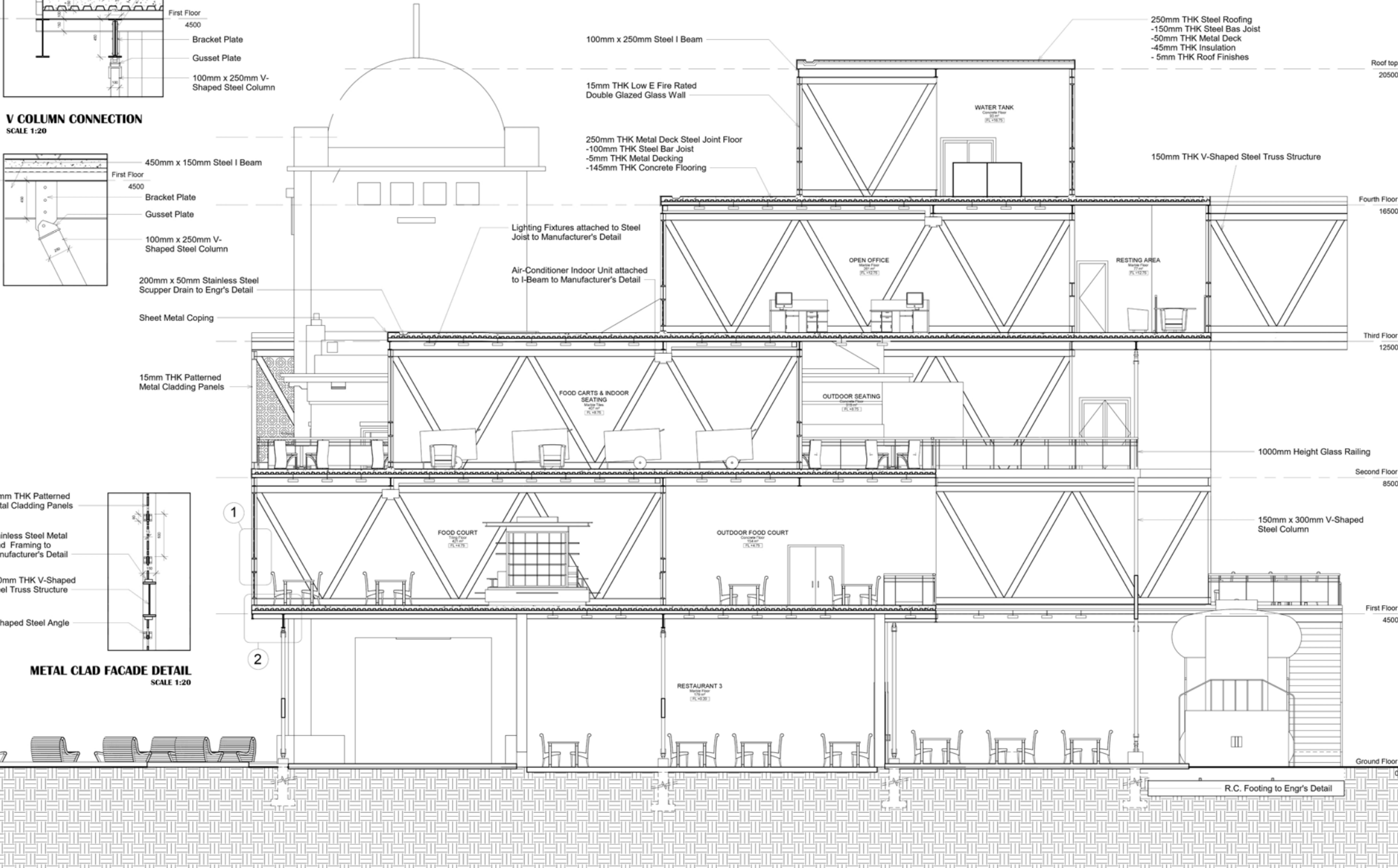


15mm THK Patterned Metal Cladding Panels
Stainless Steel Metal Clad Framing to Manufacturer's Detail

150mm THK V-Shaped Steel Truss Structure

L-shaped Steel Angle

METAL CLAD FACADE DETAIL
SCALE 1:20



TYPICAL SECTION
SCALE 1:50



OUTDOOR DINING AREA

- **Overviewing ground floor**
- **Interaction with Upper Corridor Drinking Area**



OUTDOOR PAVEMENT

- **Leading to Building Lobby**
- **Encourage walking and relaxing**
- **Appreciation of Building Facade**
- **Connected to existing Flyover from Merlin Tower**

URBAN-BRIDGE PROMENADE

NG AN QI

B15BE0069

WORKBASE DR LIM Y.L.

Food Hub Centre

The project requires exploration on local heritage in producing a community-oriented design. The building demonstrates new intervention on adaptive re-use development at Johor Bahru town. Serve as food hub centre, the building provide Food Marketing to local and tourists.



FRONT PERSPECTIVE



BIRD-EYE VIEW



NIGHT VIEW



Project number	0001
Date	10/12/2016
Drawn by	NG AN QI B15BE0069
Workbase	Dr Lim Y.L.
Scale	1 : 200

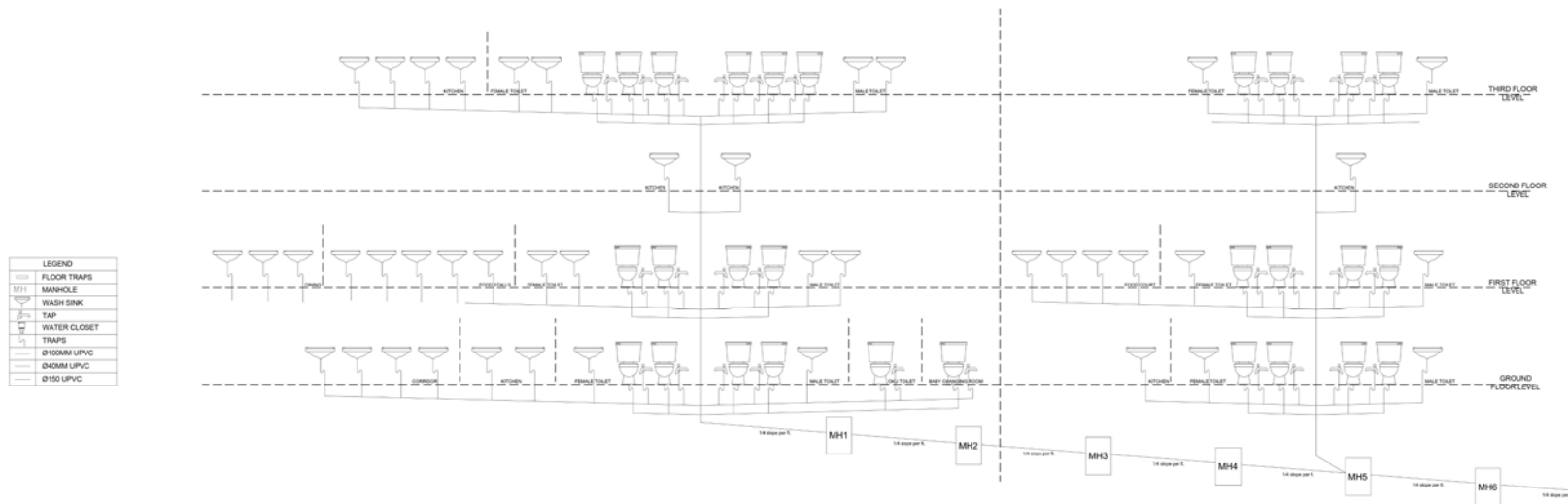
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Scale 1 : 200

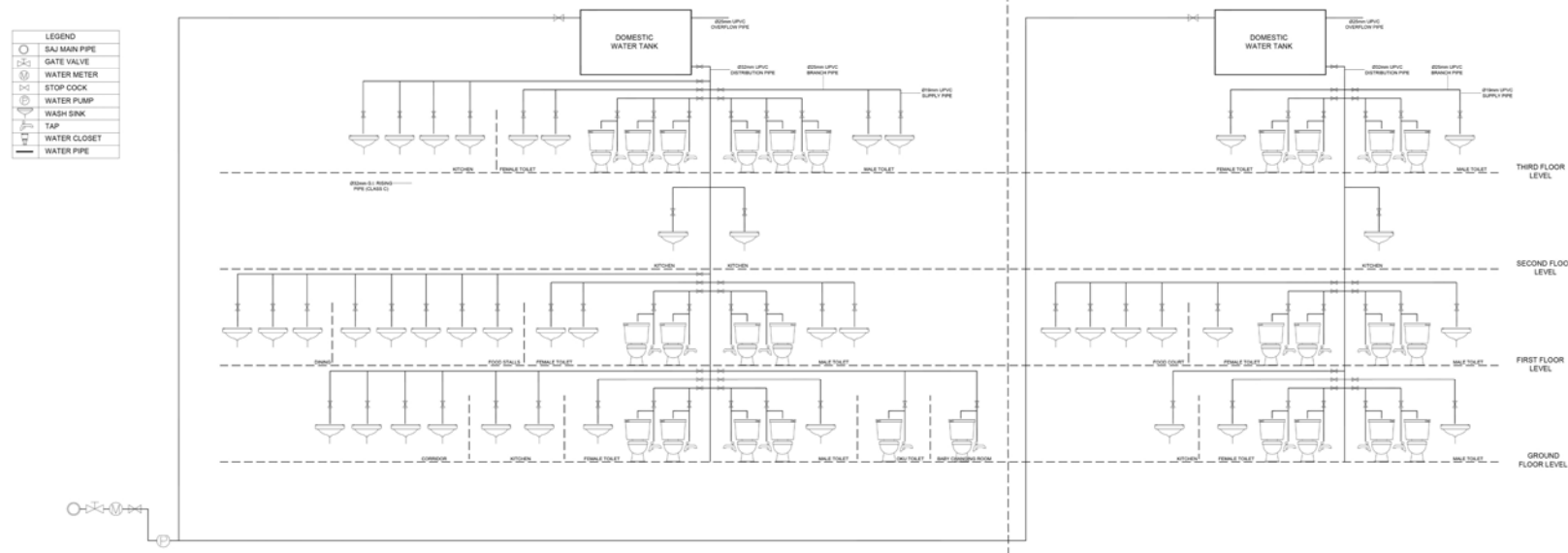


Project number	0001
Date	10/12/2016
Drawn by	NG AN QI B15BE0069
Workbase	Dr Lim Y.L.
Scale	1 : 200

SEWERAGE SECTIONAL SCHEMATIC DIAGRAM



WATER DISTRIBUTION SYSTEM



Autodesk® Revit®

SBEA 3158 DESIGN 5

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Project number	0001
Date	10/12/2016
Drawn by	NG AN QI B15BE0069
Workbase	Dr Lim Y.L.
Scale	1:50

**Life is a journey,
with problems to solve,
lessons to learn,
but most of it,
Experiences to Enjoy.**

CONTACT

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